MINI PROJECT

CAR SALES DATA ANALYSIS AND VISUALIZATION USING PYTHON, POWER BI, AND TABLEAU

Framework for Data Visual Analytics

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ABSTRACT

This mini project, carried out for the subject Framework for Data and Visual Analytics (FDVA), analyses a Car Sales dataset using Python, Microsoft Power BI and Tableau. The dataset is stored in a file named carSales.csv and contains 200 sale records with 22 columns, such as dealership state, brand, model, fuel type, transmission, customer details, ex-showroom price, discount, finance type, customer rating and sale date. The raw data had inconsistent text entries, mixed date formats, numeric values stored as text and missing values in three columns. Python with Pandas was used to correct these problems, create new columns such as Net_Sale_Price and Discount_Percentage, and save a cleaned file with 200 rows and 27 columns. Matplotlib and Seaborn were then used to create eight charts, including brand-wise sales, fuel type share, a monthly trend and a correlation heatmap. Interactive dashboards were also developed in Power BI and Tableau, with slicers for brand, fuel type and finance type, KPI cards and charts by state. The analysis showed that Hyundai had the most sales records, CNG was the most common fuel type, and the Tableau dashboard reported a total net sales value of about ₹334.77 million for all 200 sales. The project shows how preprocessing, charts and dashboards turn raw sales records into clear information that can support business decisions.

1. Data Preprocessing

The dataset (carSales.csv) has 200 rows and 22 columns. It was cleaned in Python using Pandas, as shown below.

1.1 Loading and inspecting the data

```
df = pd.read_csv("carSales.csv")
df.shape        # (200, 22)
df.info()
```

Price, discount, EMI months, waiting days and sale date were stored as text and had to be converted.

1.2 Missing values and duplicates

```
print(df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())
```

EMI_Months (24), Customer_Rating (40) and Waiting_Days (20) had missing values. No duplicate rows were found (0).

1.3 Fixing data types and filling missing values

```
df['Waiting_Days'] = pd.to_numeric(
    df['Waiting_Days'].astype(str).str.replace(
        r'\s*days?\s*$', '', regex=True, case=False),
    errors='coerce')
df['Waiting_Days'] = df['Waiting_Days'].fillna(df['Waiting_Days'].median())
# same steps for EMI_Months and Customer_Rating
```

Units such as “days” and “months” were removed, the columns were converted to numbers, and the missing values were filled with the median. After this, every column had 0 missing values.

1.4 Standardising text values

```
df['Gender'] = df['Gender'].astype(str).str.strip().str.upper()
df['Gender'] = df['Gender'].replace({'M': 'MALE', 'F': 'FEMALE'})
df['Trade_In_Vehicle'] = df['Trade_In_Vehicle'].replace(
    {'Y': 'YES', 'N': 'NO', '1': 'YES', '0': 'NO'})
df['Fuel Type'] = df['Fuel Type'].astype(str).str.strip().str.upper()
```

The same answer was written in many ways (for example M, male, MALE). These were converted to one form for Gender, Fuel_Type, Transmission, Finance_Type, Color and the Yes/No columns. Ex_Showroom_Price and Discount_INR were cleaned with a regular expression and converted to numbers.

1.5 Converting the sale date

```
df['Sale_Date'] = pd.to_datetime(
    df_original['Sale_Date'].astype(str).str.strip(),
    format='mixed', dayfirst=True, errors='coerce')
df['Sale_Year'] = df['Sale_Date'].dt.year
df['Sale_Month_Name'] = df['Sale_Date'].dt.month_name()
```

The first attempt left 154 dates missing, so the original column was parsed again with format='mixed'. After this, no date was missing (0).

1.6 New columns and export

```
df['Net_Sale_Price'] = df['Ex_Showroom_Price'] - df['Discount_INR']
df['Discount_Percentage'] = df['Discount_INR'] / df['Ex_Showroom_Price'] *
100
df.to_csv("Cleaned_Car_Sales.csv", index=False)    # shape (200, 27)
```

Five columns were added (Sale_Year, Sale_Month, Sale_Month_Name, Net_Sale_Price, Discount_Percentage). The cleaned dataset with 200 rows and 27 columns was saved and used for the visualizations and dashboards.

2. Data Visualization Using Python

Eight charts were created from the cleaned data using Pandas, Matplotlib and Seaborn (sns.set_theme(style="whitegrid")). Each chart is explained briefly below.

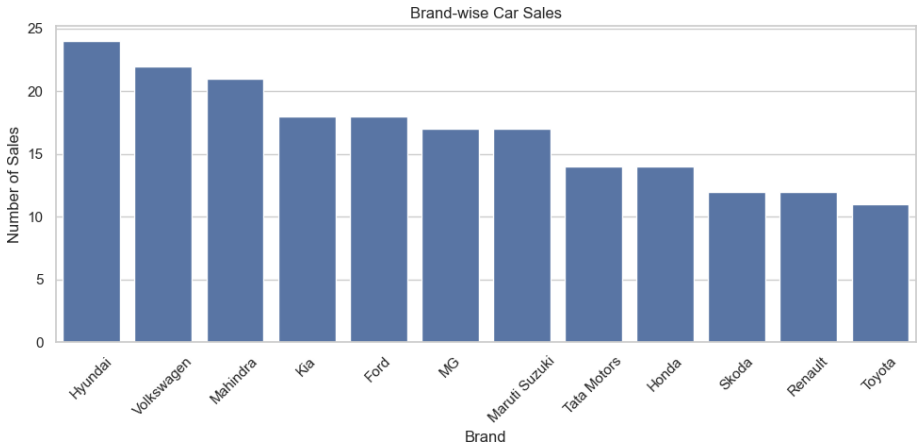


Figure 2.1: Brand-wise sales (sns.countplot)

Result: Hyundai has the most sales (24) and Toyota the fewest (11).

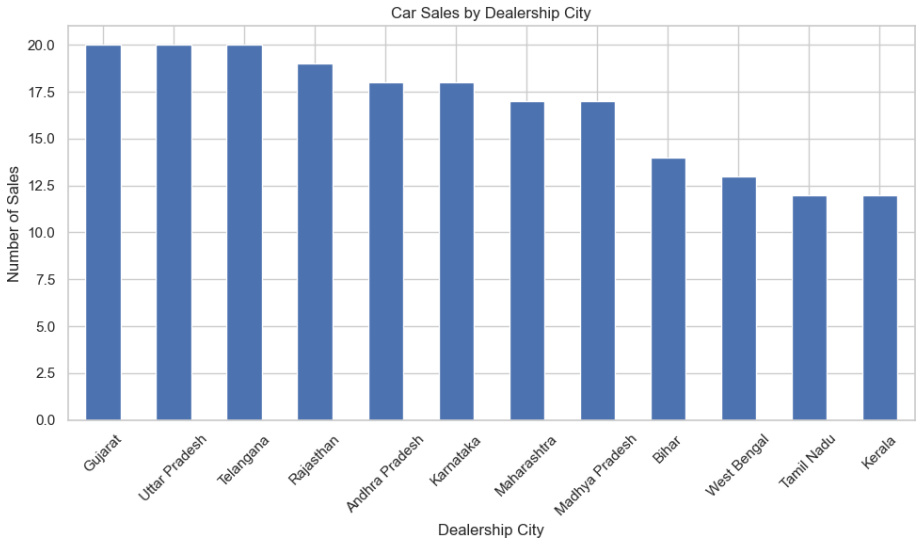


Figure 2.2: Sales by dealership state (bar chart)

Result: Gujarat, Uttar Pradesh and Telangana lead with 20 each; Kerala and Tamil Nadu have 12.

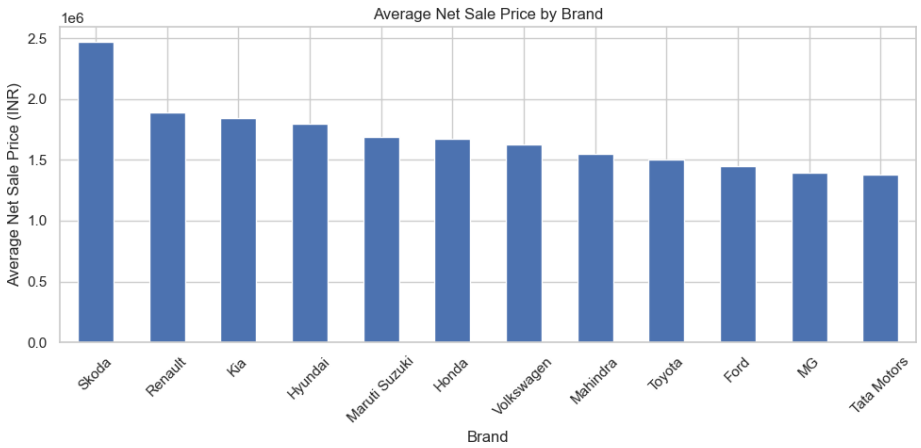


Figure 2.3: Average net sale price by brand

Result: Skoda is highest (about ₹24.7 lakh); Tata Motors is lowest (about ₹13.8 lakh).

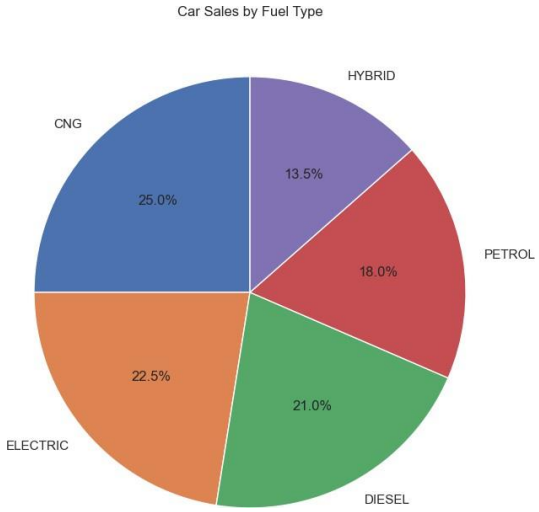


Figure 2.4: Sales by fuel type (pie chart)

Result: CNG 25%, Electric 22.5%, Diesel 21%, Petrol 18%, Hybrid 13.5%.

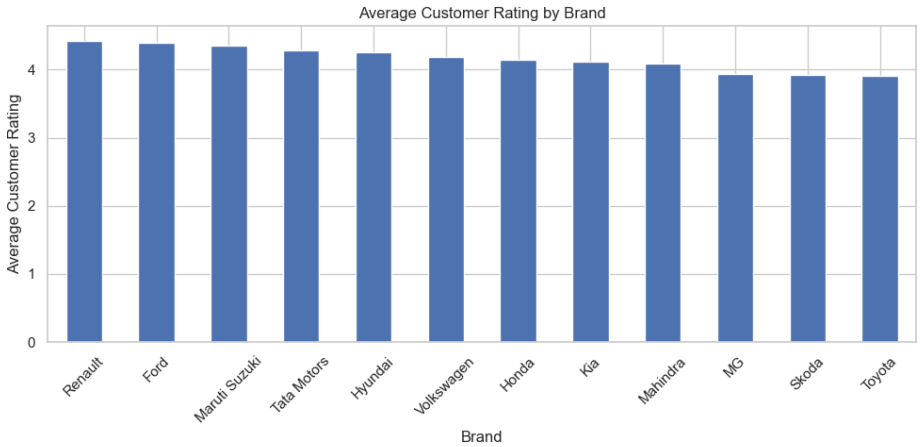


Figure 2.5: Average customer rating by brand

Result: Renault is highest (4.42) and Toyota lowest (3.91); all brands are close.

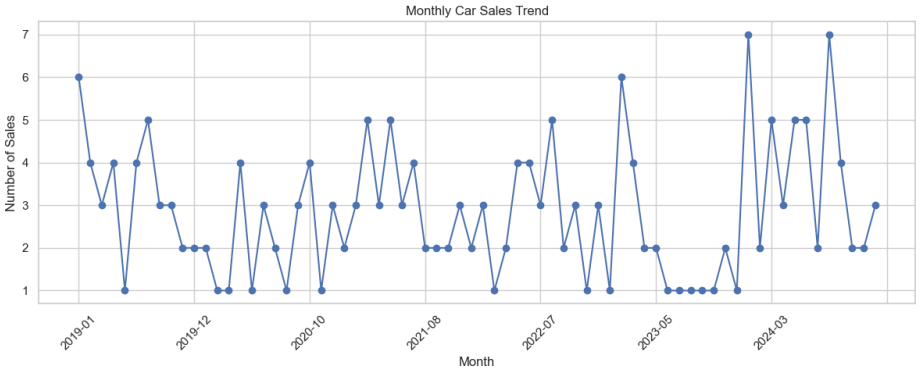


Figure 2.6: Monthly sales trend (line chart)

Result: Sales vary between 1 and 7 per month with no steady trend.

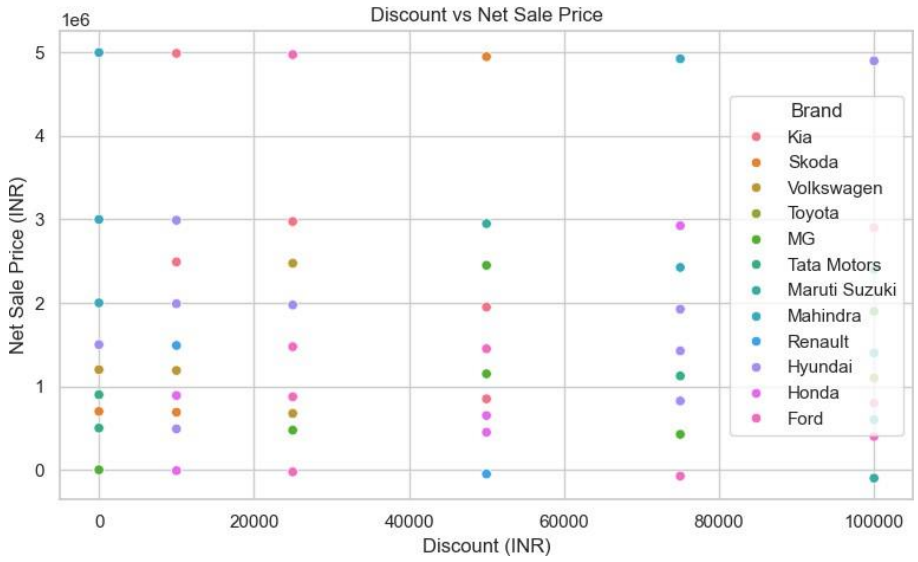


Figure 2.7: Discount vs net sale price (scatter plot)

Result: Discounts take a few fixed values and show no clear link with price.

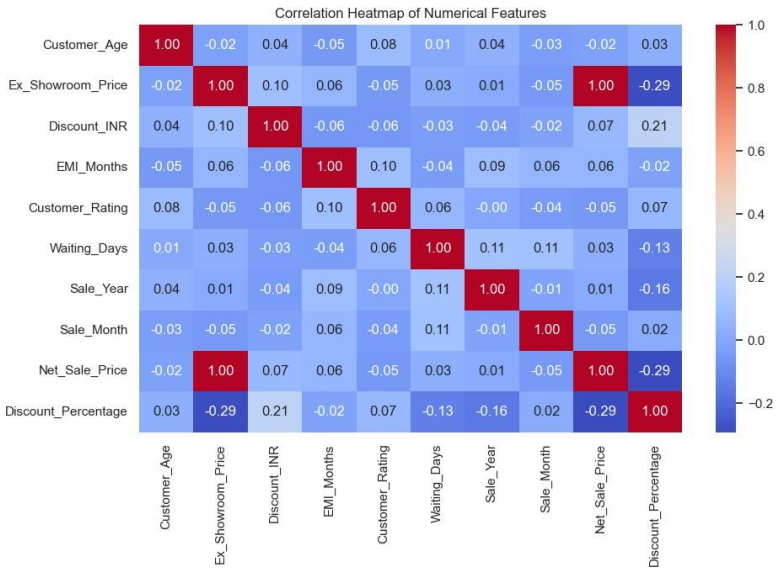


Figure 2.8: Correlation heatmap

Result: Only Ex_Showroom_Price and Net_Sale_Price are strongly related (1.00); other pairs are weak.

Overall, sales are spread fairly evenly across brands, fuel types and states, and the best-selling brand (Hyundai) is not the one with the highest price or rating.

3. Power BI Dashboard

Power BI was used to build a one-page interactive dashboard from the cleaned car sales data.

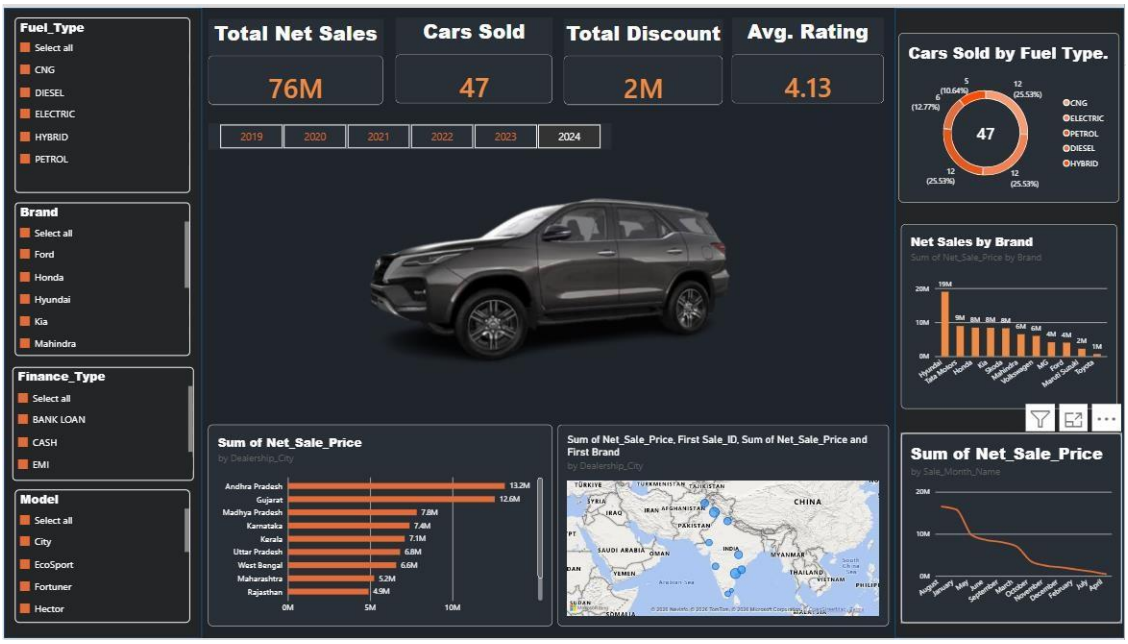


Figure 3.1: Car Sales Dashboard in Power BI

- **Slicers:** Fuel_Type, Brand, Finance_Type and Model filter the whole page; year buttons 2019–2024 are at the top (2024 appears selected).
- **KPI cards:** Total Net Sales 76M, Cars Sold 47, Total Discount 2M and Avg. Rating 4.13.
- **Charts:** a donut chart of cars sold by fuel type (47 in total), net sales by brand, net sales by dealership state, a map, and net sales by month name.
- **Insights:** Hyundai leads in net sales (about 19M); Andhra Pradesh (13.2M) and Gujarat (12.6M) lead among states; the fuel type split is fairly even.
- **Note:** the month chart is sorted by value and not in calendar order.

4. Tableau Dashboard

Tableau was used to build the "Car Sales Performance & Business Insights Dashboard" from the same cleaned data.

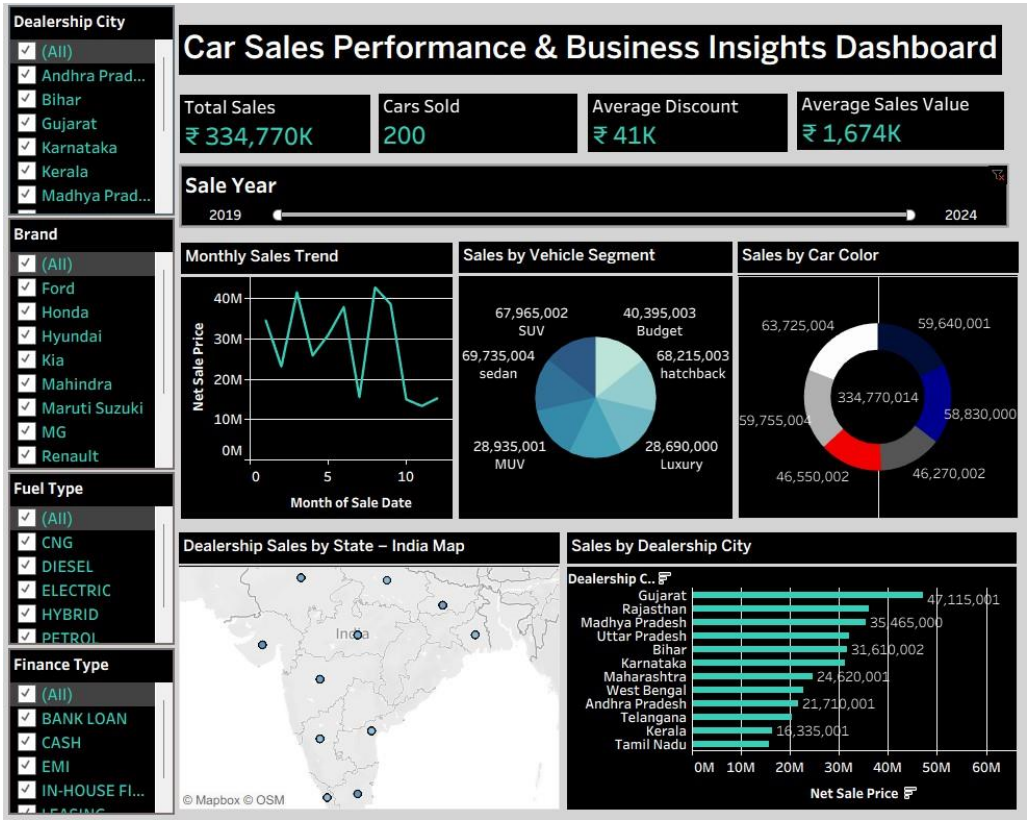


Figure 4.1: Car Sales Dashboard in Tableau

- **Filters:** Dealership City, Brand, Fuel Type and Finance Type check lists, and a Sale Year slider (2019–2024).
- **KPIs:** Total Sales ₹334,770K, Cars Sold 200, Average Discount ₹41K and Average Sales Value ₹1,674K.
- **Charts:** monthly sales trend, sales by vehicle segment (pie), sales by car color (donut), India map of dealership states, and sales by dealership state (bar).
- **Insights:** Gujarat has the highest sales (47,115,001) and Tamil Nadu and Kerala the lowest; sedan, hatchback and SUV segments are the largest; sales by colour are fairly even.